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Professional Summary

Python Software Engineer with 4+ years of production experience building reliable backend services, automating CI/CD pipelines, and owning database infrastructure end-to-end. Proven track record with PostgreSQL schema migrations, Bash automation, and AWS cloud infrastructure. Comfortable across the full delivery lifecycle — from schema design to release automation — and experienced collaborating with cross-functional engineering and data teams.

Experience

- Software Engineer – GenAI**, University of Essex — Colchester, UK Nov 2025 — Present
- Designed and implemented GitHub Actions pipelines with modular Bash scripts to orchestrate PostgreSQL schema migrations and edge function deployments across environments, cutting manual deployment time by 70% and enabling consistent weekly release cycles.
 - Built and maintained FastAPI microservices with structured CI/CD pipelines (GitHub Actions, Vercel), introducing automated health checks and rollback gates to improve uptime and deployment confidence across production services.
 - Authored and owned a suite of Bash-driven PostgreSQL migration scripts with pre/post validation and automated rollback logic, enabling zero-downtime schema changes across staging and production.
 - Engineered automated testing frameworks with comprehensive validation checks for backend code, reducing manual review time by 70% while maintaining production security and stability standards.
- Software Engineer – GenAI**, UK Data Service — Colchester, UK Jan 2023 — Oct 2025
- Designed and deployed FastAPI-based backend services on AWS (Lambda, Step Functions, EFS) with automated monitoring and retraining pipelines, reducing manual operational workload by 50% and achieving 95%+ accuracy on sensitive production datasets.
 - Architected serverless infrastructure on AWS with automated alerting and self-healing pipelines, ensuring consistent compliance and reliability across production and staging environments.
 - Implemented PostgreSQL-backed data ingestion and classification pipelines using retrieval and embedding strategies, improving metadata accuracy across millions of records in a large-scale dataset.
 - Led solution design sessions with onshore and non-technical stakeholders, translating business constraints into scalable backend architectures and delivering production-ready systems on schedule.
- Software Engineer & ML Instructor**, University of Essex — Colchester, UK Oct 2021 — Dec 2022
- Migrated MATLAB-based macroeconomic simulations to optimised C++, refactoring core data structures and algorithms to reduce simulation runtime by 8x.
 - Designed and delivered lectures and labs in Data Structures, Machine Learning, and software engineering fundamentals to 100+ students, building a reputation for clear technical communication.

Education

- University of Essex**, PhD in Computational Finance Oct 2021 — May 2025
- Built high-performance C++ data pipelines to source, process, and clean large-scale Bloomberg financial datasets, enabling AI-driven model training workflows.
 - Engineered a PyTorch-based transfer learning framework for financial time-series prediction and portfolio optimisation.
- University of Essex**, MSc in Financial Econometrics Oct 2020 — Sep 2021
- Built a C++ agent-based simulation with low-latency systems to model macroeconomic dynamics under negative interest rates.

Projects

- FinBot: Chat-Based Trading Simulation** github.com/ivan020/FinBot
- Built a real-time trading simulation in Go with microservices architecture and a PostgreSQL (Supabase) backend, serving 120+ monthly active users with reliable uptime.
 - Designed schema, migrations, and query optimisation strategies to support real-time portfolio visualisation via a JavaScript UI with interactive graphs and tables.
- Chat-LLM** github.com/ivan020/ChatLLM
- Built a Golang REST API with PostgreSQL backend for automatic card recognition and real-time price lookup, serving 50+ monthly users, with a React frontend for filtering and deck management.
- MTG Card Reader** <https://mtg-front-end-nine.vercel.app/>
- Deployed a self-hosted Ollama MCP server on Raspberry Pi 5 with custom Bash tooling and streaming website integration, enabling real-time AI-generated content delivery to live Twitch audiences.